# Assignment-3.5

TASK-1

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

| **Steps** | **What to Check** |
|---|---|
| Check the power outlet | Plug another device into the same outlet. |
| Inspect the charger and cable | Check the adapter, cable, and connector for damage or loose connections. |
| Check the charging indicator | Connect the charger and look for the laptop's charging LED. |
| Perform a power reset | Disconnect the charger and, if possible, remove the battery; hold the power button for about 10–15 seconds, then reconnect power and try again. |
| Test with a known-good compatible charger | Try another charger with the correct voltage and connector. |

TASK-2

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

1. **Check the monitor power**
   - Ensure the monitor is switched on.
   - Check the power cable and power indicator light.
   - Try another power outlet if necessary.

If the monitor has no power, fix the power connection and test again.

2. **Check the display cable**
   - Inspect the HDMI, VGA, or DisplayPort cable.
   - Ensure both ends are securely connected.
   - Try another cable if available.

If the cable is faulty or loose, reconnect or replace it.

3. **Check the monitor input source**
   - Open the monitor's menu.
   - Select the correct input, such as HDMI, VGA, or DisplayPort.
   - Check whether a "No Signal" message appears.

Select the input that matches the connected cable.

4. **Check the graphics card (GPU)**
   - Turn off and unplug the PC.
   - Check that the monitor is connected to the correct graphics output.

   If applicable, inspect the GPU seating and PCIe power connections.

A loose GPU or missing power connection may prevent the display from working.

5. **Test with another monitor**
   - Connect the PC to a known-working monitor or TV.
   - Check whether the display appears.

If the other display works, the original monitor or its cable may be faulty.

6. **Check RAM and POST indicators**
   - Turn off and unplug the PC.
   - Reseat the RAM if you are trained to do so.
   - Check motherboard diagnostic LEDs or beep codes, if available.

Faulty or improperly seated RAM can prevent the PC from completing POST.

7. **Check BIOS and software**
   - Try entering BIOS/UEFI during startup.
   - If BIOS displays correctly but Windows goes blank, investigate graphics drivers or display settings.
   - Reinstall or update the graphics driver if needed.

This helps distinguish a hardware issue from an operating-system or driver problem.

8. **Confirm the solution**
   - Restart the PC.
   - Check that the monitor displays the startup screen and Windows desktop.
   - Verify that the display remains stable.

TASK-3

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

**1.Check the mouse**

- Turn the mouse on and replace/recharge its batteries.
- Check whether the optical sensor light works.
- If it uses a USB receiver, unplug and reconnect the receiver.
- Test**:** Try the mouse on another computer. If it doesn't work there either, the mouse/receiver is likely faulty.

2. **Check the USB port**

- Plug the mouse receiver into a different USB port on the same computer.
- Test another USB device in the original port.
- Result**:** If another device works but the mouse receiver doesn't, the receiver/mouse is more likely the problem. If multiple devices fail in that port, the USB port may be faulty.

3. **Check wireless pairing**

- If it is Bluetooth, make sure Bluetooth is enabled and the mouse is paired.
- Remove the mouse from the Bluetooth device list and pair it again.
- For a 2.4 GHz mouse, check that the receiver is the correct one and is properly connected.

4. **Check system settings**

- Open Settings → Bluetooth & devices → Mouse and confirm the mouse is detected.

- Open Device Manager → Mice and other pointing devices and check for warning symbols.
- If necessary, update/reinstall the mouse driver.

5. **Confirm the cause**
   - Test the mouse on another USB port and another computer.
   - Works on another computer → check the original computer's USB port, Bluetooth settings, or drivers.
   - Doesn't work anywhere → the mouse or wireless receiver is likely faulty.

TASK-4

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

**1. Faulty or slow cooling fan:** Check whether the CPU/GPU/case fans are spinning properly when the PC is running. A fan that is stopped, noisy, or spinning slowly may not provide enough cooling.

**2. Dust buildup:** Turn off and unplug the PC, then inspect the fans, heatsinks, and air vents for dust. Clean accumulated dust carefully using appropriate compressed air.

**3. Dried or improperly applied thermal paste:** Check the contact between the CPU/GPU heatsink and processor. If the thermal paste is dried out or poorly applied, clean the old paste and apply fresh thermal paste before reinstalling the heatsink.